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Roll No

FT-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)
Operations Research and Supply Chain

Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. a) Discuss briefly the steps to formulate linear programming problem. Explain with a suitable Example.
b) Write a method of solving an assignment problem, Illustrate with an example.
2. a) Discuss the four fundamentals of supply chain in brief.
b) Explain the factors influencing the bullwhip effect and the ways to minimize the bullwhip effect.
3. a) Derive the EOQ formula for the manufacturing model without shortages.
b) What are the types of inventory? Why they are maintained? Explain the various cost related to inventory.
4. a) Define M/M/1 queuing system and state when such a system is to be considered.
b) What do you understand by rationality in game? Discuss the concept of Optimal Response Curves.

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PTO

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5. a) Describe some methods, which are useful for decision making under uncertainty. Illustrate one of them by an example.
- b) Explain clearly the various ingredients of a decision problem. What are the basic steps of a decision making process?
6. Find the initial basic feasible solution to the transportation problem shown below, by;
- Minimum cost method, and
 - North-west corner rule, State which of the method is better.

		To:			Supply
		P	Q	R	
From:	A	2	7	4	5
	B	3	3	1	8
	C	5	4	7	7
	D	1	6	2	14
	Demand	7	9	18	34

7. At a garage, car owners arrive at the rate of 6 per hour and are served at the rate of 8 per hour. It is assumed that the arrival follows Poisson's distribution and the service pattern is exponentially distributed. Determine average queue length and average waiting time.
8. Write short notes on followings(any four) :
- Saddle point
 - Monte Carlo simulation
 - Multiple server model
 - Hurwicz criteria
 - Vogel's method
 - MRP calculation

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